

Near-infrared light therapy for recovery of cerebral hypoperfusion induced by bilateral common carotid artery stenosis in mice

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ABSTRACT

Low-level light therapy, which is a red or near-infrared light-employed therapeutic methodology, can act to help the brain repair in cases of traumatic brain injury and stroke. In this paper, we investigate the effects of near-infrared light therapy (NILT) for the recovery of blood flow of mice with cerebral hypoperfusion, which is a key mechanism leading to vascular dementia, induced by bilateral common carotid artery stenosis (BCAS). The mice are divided into three groups of 4-5 mice per group: a normal group (without BCAS operation), a BCAS group (without NILT), or a BCAS/NILT group. The LED with a peak wavelength of 810 nm and a power of 20 mW is arranged to illuminate on the top of the mouse head. The mice receive treatment from the LED source 3 times per week for one month. After the treatment, positron emission tomography is used to quantify the effects of NILT on whole brain and regional cerebral blood flow in the cortex, striatum, and hippocampus.

Keywords: Bilateral common carotid artery stenosis, Near-infrared light therapy, Positron emission tomography

1. INTRODUCTION

Low-level light therapy (LLLT) has been widely used as powerful modality for the non-invasive treatment with the functions of stimulation of healing, protection of tissue from dying, increase of mitochondrial function, and improvement of blood flow and tissue oxygenation [1, 2]. LLLT can also act to help the brain repair in cases of traumatic brain injury (TBI) and stroke by stimulating neurological function, increasing memory and learning capacity, and reducing inflammation and cell death in the brain [3-5].

Red and near-infrared (NIR) light, which can penetrate into the scalp and skull, is absorbed by two different chromophores. The red and NIR light with wavelength range of 600-940 nm is primarily absorbed by the enzyme cytochrome c oxidase (CCO), which is a complex protein functioning as unit IV in the mitochondrial respiratory chain of the cortical neurons. The NIR light with wavelengths of 980 and 1064 nm is absorbed by transient receptor potential (TRP) calcium channels. The absorbed photons in both cases upregulate the cell signaling and messenger molecules including reactive oxygen species (ROS) and adenosine triphosphate (ATP), leading to activation of transcription factors [6]. These transcription factors can induce the increase of gene expression linked to protein synthesis, cell migration and proliferation, anti-inflammatory signaling, anti-apoptotic proteins, and antioxidant enzymes.

The control of cerebral blood flow, which ensures proper delivery of oxygen (O₂) and glucose to neurons, is critically important for healthy brain function and viability [7]. The insufficient cerebral blood flow (hypoperfusion) can produce white matter damage, which can cause stroke and vascular dementia, which is the second most common dementia after Alzheimer's disease [8, 9]. Cognitive impairment, which is common in people with vascular dementia, is often associated with significant atrophy of hippocampal lesions [10]. Thus, the cerebral blood flow in hippocampus is a key factor to elucidate the pathophysiology of vascular dementia.

In this study, the effects of near-infrared light therapy (NILT) is tested in a mouse model with cerebral hypoperfusion induced by bilateral common carotid artery stenosis (BCAS). The custom-made NILT system have a peak wavelength of 810 nm and a power of 20 mW. After the NILT for one month, the cerebral blood flow is measured by positron emission tomography (PET) scans in the region of cortex, striatum, and hippocampus.

2. MATERIALS AND METHODS

2.1 Subjects

Adult male C57BL/6 mice (8 weeks, 22-26 g) were used in this study. The mice were housed at one mouse per cage and were maintained on a 12-hour light/dark cycle with free access to food and water. The mice were adjusted to these conditions for 1-2 weeks prior to the experiments. The mice were assigned to the following three groups of 4-5 mice per group: a normal group (without BCAS operation), a BCAS group (without NILT), or a BCAS/NILT group. All animal care and experimental use were in accordance with the Osong Medical Innovation Foundation guidelines.

2.2 Experimental model of BCAS

Chronic cerebral hypoperfusion was induced via bilateral common carotid artery stenosis, as previously described [11]. Briefly, mice were anesthetized using the combination of zoletil (30 mg/kg) and xylazine (10 mg/kg) via intraperitoneal injection. The carotid arteries were exposed and microcoils (inner diameter 0.18 mm) were applied in the bilateral carotid arteries.

2.3 Near-infrared light therapy

Mice received the treatment using a custom-made NILT system composed of power controller, micro controller unit (MCU), LED controller, and LED modules. For NILT, LED modules with a peak wavelength of 810 nm and a power of 20 mW were placed on the top of the mouse head. The mice received treatment from the LED source 3 times per week for one month.

2.4 Positron Emission Tomography

All positron emission tomography (PET) scans were acquired using a Mediso nanoscan PET/CT (PET/SPECT-CT, Mediso Ltd., Budapest, Hungary). After injection of ^{18}F -Fluorodeoxyglucose (^{18}F -FDG) through the tail vein, the temporal change of ^{18}F -FDG uptake was monitored during first 5 min and between 45 and 90 min of PET scan. The mice were kept in the scanner under anesthesia between scans. The standardized uptake value ratio (SUVr) was obtained using the decay-corrected amount in injected ^{18}F -FDG and the weight the mouse into account.

3. RESULTS AND DISCUSSION

Figure 1 shows the custom-made NILT system composed of power controller, MCU, LED controller, and LED modules. The LED modules have a peak wavelength of 810 nm and a power of 20 mW. To efficiently illuminate the NIR light on the mouse head, each mouse was placed inside a plastic restrainer and the LED module was attached on the mouse head. The dimmer was used for controlling the LED intensity.

To investigate the NILT treatment, mice were divided into three groups: Group I (n=5), normal group without BCAS operation and NILT treatment; Group II (n=4), BCAS group with BCAS operation and without NILT treatment; Group III (n=4), BCAS/NILT group with BCAS operation and NILT treatment as shown in Fig. 2. Male C57BL/6 mice (8 weeks, 22-26 g) were anesthetized with the combination of zoletil (30 mg/kg) and xylazine (10 mg/kg) and chronic cerebral hypoperfusion was induced by applying microcoils with inner diameter of 0.18 mm to both common carotid arteries. After the operation, the mice were housed in cages with a 12-hour light/dark cycle. At 1 week after BCAS operation, the NILT treatment was performed 3 times per week for one month. After then, the PET scans were conducted to indirectly disclose the effect of NILT treatment on cerebral blood flow.

Figure 3 shows the SUVr after BCAS operation observed in cortex, striatum, and hippocampus. The SUVr gradually decreased for 20 weeks after BCAS operation, indicating that the chronic cerebral hypoperfusion was successfully induced to the mice through BCAS operation.

Figure 4 shows the summed ^{18}F -FDG PET images in the coronal plane obtained from the 0-5 min of the scans of in the normal, BCAS, and BCAS/NILT group. The images are superimposed on a standard MRI atlas.

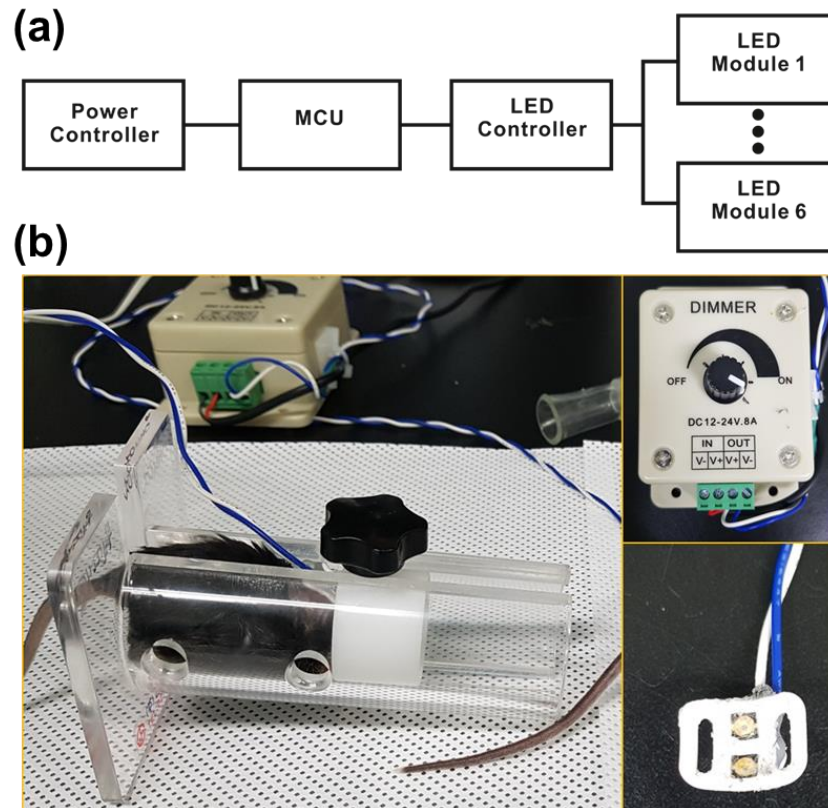


Figure 1. Custom-made NILT system. (a) The NILT system was composed of power controller, micro controller unit (MCU), LED controller, and LED modules. LED modules with a peak wavelength of 810 nm and a power of 20 mW were placed on the top of the mouse head. (b) To efficiently illuminate NIR light on the mouse head, each mouse was placed inside a plastic restrainer. The dimmer was used for controlling the LED intensity.

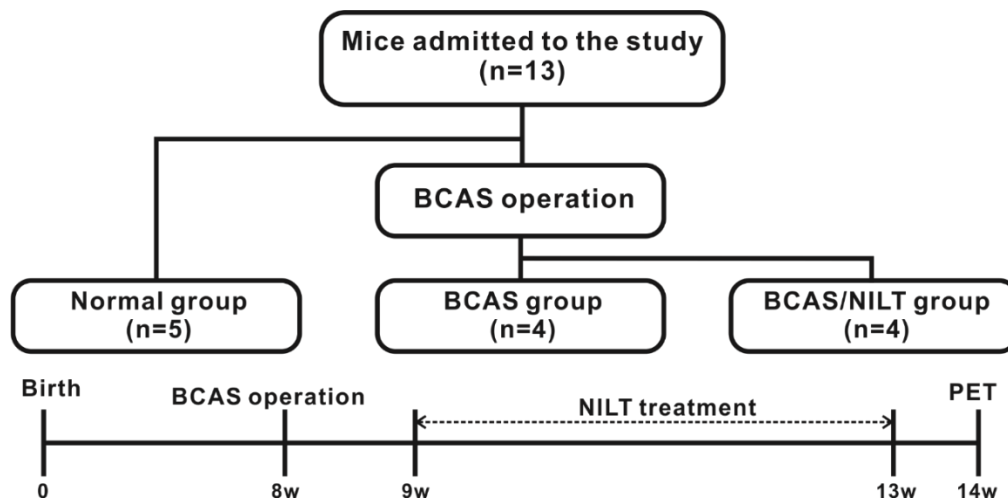


Figure 2. Experimental design for NILT. (a) The mice were divided into three groups of 4-5 mice per group: a normal group (without BCAS operation), a BCAS group (without NILT), or a BCAS/NILT group.

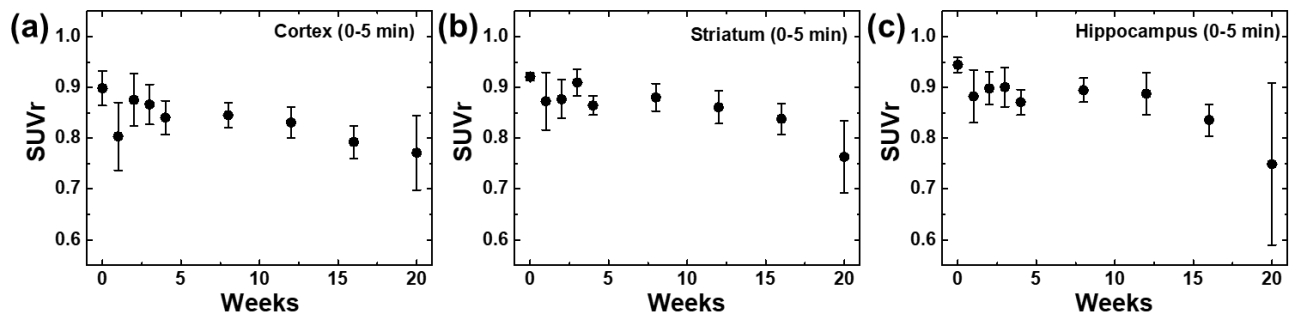


Figure 3. SUVR vs. time of the BCAS group in the region of cortex, striatum, and hippocampus.

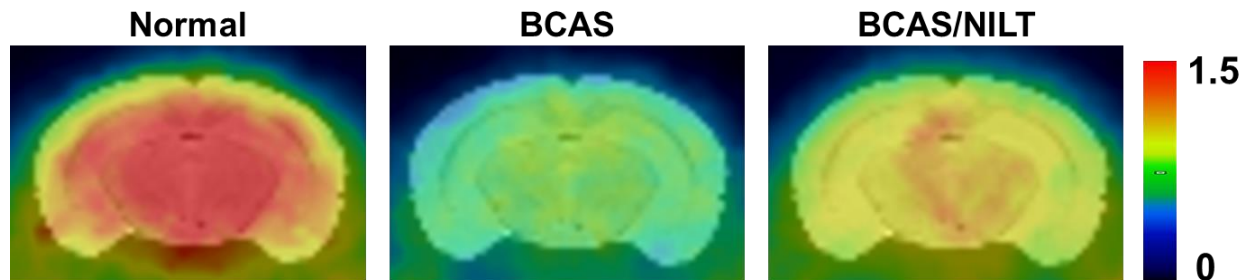


Figure 4. PET images in the coronal plane obtained from the 0-5 min of the scans in the normal, BCAS, and BCAS/NILT groups. The color scale bar shows a SUVR of 0-1.5.

4. CONCLUSION

We demonstrated the custom-made NILT system for brain repair of the mouse with vascular dementia. To efficiently deliver light energy, the curved LED module fitted to the mouse head was presented. The chronic cerebral hypoperfusion, which can cause vascular dementia, was successfully induced to the mice through BCAS operation. In the near future, we would like to investigate the effects of NILT for the recovery of blood flow in the mice with cerebral hypoperfusion.

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